

IT462 — Exploratory Data Analysis — Lab 8

A Complete EDA Workflow on a Dataset of Your Choice

General Instructions

- Use Python with pandas, numpy, matplotlib, and seaborn (or plotly).
- Answer all questions in a single Jupyter Notebook, in the given order.
- Follow each code cell with a short markdown cell containing your observation.
- All plots must have a title, labeled axes, and a legend where applicable.
- State any assumptions made while handling missing values or outliers.
- Mention the dataset name and source link in the first cell.
- Rename your notebook as RollNumber_Name_Lab8.ipynb before submission.

Dataset Requirements

Select a dataset from a public source (Kaggle, UCI, data.gov.in, etc.). Toy datasets such as Iris, Titanic, and Boston Housing are not permitted. The dataset must have:

- At least 500 rows, 6 numerical columns, and 2 categorical columns.
- A target variable (binary/multi-class label or numeric).
- Missing values in at least 3 columns and outliers in at least 3 numerical columns. If absent, introduce them artificially and document the changes.

In the questions below, “chosen column” and “target” refer to columns of your dataset.

Question 1 | Data Loading and Initial Exploration

- (a) Load the dataset and display the first 10 and last 5 rows. Identify the target variable.
- (b) Report the shape, and the number of numerical and categorical columns.
- (c) Use info() to check data types and non-null counts. Correct any wrong data types.
- (d) Use describe() for numerical and categorical columns. Identify three suspicious columns using specific values.
- (e) Report and remove duplicate rows, if any.
- (f) Show value_counts for categorical columns and comment on imbalance or inconsistent labels.

Question 2 | Missing Value Handling

- (a) Tabulate missing values per column (count and percentage), sorted in descending order.
- (b) Visualize the missing-value pattern and write two observations.
- (c) For three columns, state whether the missingness is MCAR, MAR, or MNAR, with justification.
- (d) Treat missing values using: (i) deletion, (ii) mean/median imputation, (iii) categorical imputation, and (iv) one advanced method (group-wise, KNN, or regression). Justify each choice.
- (e) Verify that no missing values remain, and compare the mean and standard deviation of one column before and after imputation.

Question 3 | Outlier Detection and Treatment

- (a) Plot box plots for all numerical columns in a grid, and histograms/KDE plots for four columns. Comment on skewness.

- (b) Apply the IQR method to four columns. Report the bounds and outlier count for each.
- (c) Apply the Z-score method to the same columns. Compare the counts with IQR and explain the differences.
- (d) Draw a scatter plot of two numerical columns, highlighting the outliers.
- (e) Treat outliers using: (i) removal, (ii) capping/winsorization, and (iii) a log or square-root transformation (show before and after histograms). Justify each choice.
- (f) Regenerate box plots for the treated columns and summarize the changes.

Question 4 | Correlation Analysis

- (a) Display the Pearson correlation matrix as an annotated heatmap.
- (b) Identify the strongest positive and strongest negative pairs, and two near-zero pairs. Explain each briefly.
- (c) Create a pair plot of five numerical columns coloured by the target (bin the target if numeric). Note two patterns.
- (d) Compute and rank each feature's correlation with the target. Identify the most useful predictors.
- (e) Compare Pearson and Spearman matrices, and explain any notable differences.

Question 5 | Feature Engineering

- (a) **Binning:** Bin one continuous column into meaningful categories (e.g., age groups) and show the count in each bin.
- (b) **Ratio feature:** Create a new feature by combining two columns (e.g., $A / (B + 1)$). Explain what insight it captures.
- (c) **Encoding:** Encode categorical columns using an appropriate method for each (One-Hot for nominal, Ordinal/Label for ordered). Justify two choices.
- (d) **Flag feature:** Create a binary feature using a threshold on a numerical column (e.g., above the 75th percentile). State how you chose the threshold.
- (e) **Scaling:** Apply Min-Max and Standardization to one column. Show summary statistics before and after, and state when each is preferred.
- (f) **Visualization:** Create one final plot summarizing an insight about the target, and justify the plot type.
- (g) **Report:** Write a 250–400 word summary covering data quality issues, missing value and outlier handling, key correlations, and the most useful engineered features.

End of Lab 8 — Ensure your notebook runs top to bottom without errors before submission.